

EXAM 7 – SPRING 2014

17. (1.5 points)

Given the following financial data for an insurer:

	2014	2015	2016
Beginning U.S. GAAP Equity	\$3,000,000		
Forecasted Net Income	\$420,000	\$500,000	\$550,000

In order for the insurer to maintain its A+ Rating, it must meet a minimum capital requirement for each year equal to 5% above the beginning U.S. GAAP equity for the given year.

a. (1 point)

Calculate the Free Cash Flow to Equity (FCFE) for 2016.

b. (0.5 point)

Explain why the FCFE valuation is unaffected by whether the insurer's loss and loss adjustment expense reserves increased in 2014.

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 17				
TOTAL POINT VALUE: 1.5			LEARNING OBJECTIVE: B1: Calculate the effect of loss and expense reserve requirements and regulatory or rating agency capital requirements on the free cash flow to equity for a P&C insurer	
SAMPLE ANSWERS				
Part a: 1 point				
<i>Sample Answer 1</i>				
		2014	2015	2016
(1)	Beg Equity	3,000,000	3,150,000	3,307,500
(2)	Rqd Capital	3,150,000	3,307,500	3,472,875
(3)	Net Income	420,000	500,000	550,000
FCFE=(3)-[(2)-(1)]		270,000	342,500	384,625
For 2016 Beg Equity =3,000,000 × (1.05) ² =3,307,500				
Increase in Required Capital=3,307,500 × (.05)=165,375				
FCFE=Net Income - Increase in Capital=550,000-165,375=384,625				
<i>Sample Answer 2</i>				
Yr	Beg Req(000s)	Req Cap↑ (000s)	NI(000s)	FCFE=NI-↑Cap
14	3,000			
15	3,150			
16	3,307.5	165.375	550	384.625
17	3,472.875			
Part b: 0.5 point				
<i>Sample Answer 1</i>				
The formula for Free Cash Flow to Equity (FCFE) = Net Income + Non-cash Charges – Net Working Capital (NWC)I –Increase in Required Capital +Net Borrowing				
The reserves would appear in both the Non-cash charges and Increase in Required Capital. These are offsetting, so the loss and LAE reserves do not impact the FCFE.				
<i>Sample Answer 2</i>				
The change in loss and LAE reserves are accounted for twice. It is accounted for in the Net Income originally and then in the capital requirement increase, which both offset each other. Thus changes in reserves do not affect FCFE.				
EXAMINER’S REPORT				

SAMPLE ANSWERS AND EXAMINER'S REPORT

Part a

The candidate is expected to know Free Cash Flow to Equity for a P&C insurer.

To get full credit, the candidate has to assume all free cash flow is paid to shareholders. In general, candidates performed well in this part with 78% receiving full credit. Some candidates calculated beginning equity based on previous years beginning equity plus net income.

Part b

The candidate is expected to know FCFE formula and show how Loss and LAE reserves are reflected in the formula.

To get full credit, the candidate has to mention the offsetting effect in different parts of the formula. Over half of the candidates received full credit. Some candidates misinterpreted the question and stated that Net income for 2016 is forecasted so reserve change in 2014 has no impact.

Errors included referencing that increase in reserves only affected certain calendar years, referencing reserves as debt, referencing reserves being charged twice in the FCFE formula, and determining net income was based on a forecast.